



Perceived value and its impact on choice behavior in a retail setting

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Abstract

Value has been widely recognized in the past decade by retailers as an important consideration in consumers' store selection and product purchase decisions. Empirical academic research, however, has concentrated on the impact of consumers' perceived value on the *attitude* to purchase rather than on the resulting behavior itself. We address the substantive retailing issue of whether consumers' value orientation (i.e., their predisposition towards quality, price or a balance of the two) actually affects their choice behavior in-store, for the case of durable goods. This paper discusses an approach to modeling consumer choice behavior (buy/not buy given the store) based on consumers' value orientation and their perceptions of various product and store characteristics appropriate to the specific in-store experience. The consumer market is assumed to consist of several unobservable segments, membership of which reflects a consumer's general value orientation concerning durable goods. The utility of buying the specific product in question from a certain retailer is conditional on the consumer's segment membership. From a modeling viewpoint, the proposed framework requires the joint estimation of a latent segment membership function (based on value orientation) and a discrete choice model. The approach we adopt requires the formulation of an *ordered latent segment choice model*, since *a priori* considerations permit us to predict the number and structure of the segments. The empirical results presented in this article are based on a survey of over 1000 shoppers actively searching for an electrical appliance in various retail stores. © 2000 Elsevier Science Ltd. All rights reserved.

Keywords: Perceived value; Retail choice behavior; Choice modeling; Latent segmentation

1. Introduction

While the importance of perceived value is recognized in terms of its expected positive outcomes on sales, margins, revenue and market share, previous empirical studies have investigated the effect of value on perceptual outcomes, usually willingness-to-buy (Baker, 1990; Dodds et al., 1991). No research has been conducted to assess behavioral outcomes of perceived value, despite the conceptualization by Zeithaml (1988) in a landmark paper modeling purchase as the outcome of perceived value.

The present research study investigates the link between perceived value and behavior, and is important for two reasons. Firstly, it is clear that understanding the impact of perceived value on consumer behavior (as opposed to perceptions) is more directly relevant to outcomes such as sales and revenue, which are of salient

interest to retail and other organizations. Secondly, the well established fact that behavioral intentions are relatively poor predictors of behavior (Juster, 1966; Pickering and Isherwood, 1974; Morwitz and Schmittlein, 1992) further emphasizes the importance of direct evaluation and study of the behavioral outcomes of perceived value rather than proxy measures such as willingness-to-buy.

In this research we conceptualize a model of the effect of perceived value on choice (i.e., buy/not buy given the store, so that store selection is not part of the choice process modeled here). Consumers are divided into segments according to their general predisposition towards value, that is, their "value orientation". This orientation is conceptualized as the attitude towards price and quality in any consumption experience for a given class of product (e.g., purchase of a durable). Each such segment is proposed to differ in terms of the relative importance placed on various characteristics of the purchase situation (e.g., store and product characteristics). Associated with the conceptual framework is an econometric model that simultaneously classifies consumers into the segments and predicts their choice behavior conditional on segment membership (see also Swait, 1994; Gopinath and

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Ben-Akiva, 1995). A structural model recognizing the conditionality of preferences on segment membership should produce market segmentation and choice predictions that are both more actionable and directly relevant to decision making. In addition, our theoretical development leads us to expect that the latent segments will maintain a certain ordinal relationship between themselves, in terms of an underlying latent dimension. This leads to the formulation and estimation of an *ordered latent segment model* (see also Gopinath and Ben-Akiva, 1995).

Thus, a major objective of this research study is to determine whether consumers' value orientation (i.e., affinity towards price, quality or a balance of the two) is a useful basis for segmentation of the marketplace. And if so, we wish to determine the extent to which this orientation results in a differential impact of various variables (such as the product's price) on consumers' product utility assessments. It is also of methodological interest to test the proposed econometric modeling approach.

We, therefore, structure the paper as follows: the next section reviews the perceived value literature and introduces the concept of value orientation, which serves as the basis for development of a theoretical market segmentation framework and a supporting modeling approach. We then present the application and discuss the empirical validation of the proposed segmentation model. The paper concludes by summarizing our findings, discussing their implications and suggesting avenues for further research.

2. Value orientation as a basis for market segmentation

In this section we review the marketing literature on the perceived value construct. We then introduce the concept of "value orientation" and suggest its use as the basis for introducing value as a market segmentation variable. The resultant conceptual framework leads to the formulation of several propositions concerning segment structure, which are developed hand in hand with a specific ordered latent segment model.

2.1. Previous research on the perceived value construct

Following Zeithaml's (1988) conceptual model of value, empirical research modeling the role of value in the consumer decision making process has been conducted by Baker (1990), Dodds et al. (1991) and Sweeney et al. (1997). While the former two works used an experimental approach on student subjects, the latter surveyed consumers making decisions in stores. Kerin et al. (1992) modeled the effect of various antecedents of value, but not its outcomes. All these studies included price and product quality as antecedents of value. However, Baker (1990), Kerin et al. (1992) and Sweeney et al. (1997)

included both product and service quality, while store image and brand image were also included by Dodds et al. (1991). In all the above studies in which the outcome of value was measured, it was conceptualized as "willingness-to-buy" the product rather than the purchase behavior itself. In these research studies, value was found to have a significant positive effect on the perceptual variable "willingness-to-buy". In no case was the outcome of value measured in terms of actual behavior, as we do in this paper.

Value is clearly an important variable in the purchase decision process (e.g. Dodds et al., 1991; Sweeney et al., 1997). However, consumers will differ in terms of the importance they place on quality and price when purchasing durable goods. We argue below that this categorization can serve as the basis for segmentation of the market place.

2.2. Value orientation: the basis for market segmentation

From a marketing perspective, although individuals may have unique shopping attitudes and perceptions, it is clearly useful and more cost effective to group consumers into classes so that specific marketing strategies can be used to address different segments. Previous research has indicated that consumers vary in their personal sensitivity or orientation towards product characteristics. Consumers have been evaluated, for example, in terms of their ethnocentrism (Shimp and Sharma, 1987), energy consciousness (Lumpkin, 1985), fashion consciousness (e.g., Wilkes, 1992) and value consciousness (Lichtenstein et al., 1990).¹ In this research we conceptualize a choice model in which consumers are divided into segments according to their predisposition towards value. This will be termed their "value orientation". For the purposes of this paper, this "value orientation" can be thought of as the innate attitude towards obtaining a low price, high quality or a combination of quality and price in a consumer's purchase of a specific product class (e.g., purchase of a durable good). Given that segments differ in terms of general tendencies towards value in the purchase of the product class, each segment is then proposed to differ in terms of the relative importance or utility placed on various characteristics relating to a specific incident in the retail environment. For example, it is postulated that a consumer's value orientation underlies his or her price sensitivity in a specific situation, which translates into consequent utility differences.

To be specific, assume that a population of consumers has S latent (i.e., unknown to the analyst) segments. Each

¹ It should be noted that Lichtenstein et al. (1990) explored the distinction between coupon proneness and value consciousness with respect to coupon redemption behavior and did not pursue the same line of segmentation research as we do.

segment has associated with it the taste parameter vector β_s , $s = 1, \dots, S$, which we postulate forms the basis for their evaluation of the attractiveness of a product offering. Thus, we assume that the utility $U_{i|s}$ of the i th product offering, conditional on a consumer's membership in segment s , is given by

$$U_{i|s} = \beta_s X_i + \varepsilon_{i|s}, \quad (1)$$

where X_i is the vector of product attributes, personal characteristics and context descriptors, and $\varepsilon_{i|s}$ is an error term. If we assume that the distribution of the error terms over alternatives, conditional on segment membership, is IID Gumbel with scale factor μ_s , then the conditional choice probability of choosing an alternative $i \in C$ (the choice set) is the familiar Multinomial Logit (MNL) model:

$$P_{i|s} = \exp(\mu_s \beta_s X_i) / \sum_{j \in C} \exp(\mu_s \beta_s X_j). \quad (2)$$

When we observe a randomly selected individual from the population, we are unable to observe their segment membership. Instead, suppose we observe *value orientation* VO_n , based on which we define Y_n , a latent segment membership scoring function, as follows:

$$Y_n = \alpha VO_n + \delta(VO_n)(DE_n) + v_n, \quad (3)$$

where α, δ are parameters, DE_n is the consumers self-perception of his or her difficulty to evaluate the quality of the good and v_n is an error term. Since the consumer's difficulty in evaluating the quality might decrease the confidence in the use of quality perceptions, it is likely that less confident consumers would rely more on price and less on quality. This follows from the evidence that extrinsic cues are more often used than intrinsic cues when knowledge of the product category is low and quality is difficult to evaluate (Zeithaml, 1988; Rao and Monroe, 1988). This leads us to suppose that the effect of value orientation (i.e., the extent to which consumers focus on quality and/or price) should be moderated by the consumer's perceived ability to evaluate quality; specifically, it is expected that $\alpha > 0$ and $\delta < 0$.

Define a latent segment membership indicator I_n , which takes on values in the set $\{1, \dots, S\}$. It is related to Y_n as follows:

$$I_n = \begin{cases} 1 & Y_n \leq \tau_1, \\ 2 & \tau_1 \leq Y_n \leq \tau_2, \\ \vdots & \vdots \\ S & \tau_{S-1} \leq Y_n, \end{cases} \quad (4)$$

where τ_s , $s = 1, \dots, S-1$, are cutoff parameters to be estimated that define the ranges of Y_n that lead to classification into each latent class. (Note that only $S-1$

cutoff parameters are needed to construct S classes.) Expression (4) imposes an ordinal relationship among the latent segments: membership in higher-order segments implies higher values of Y_n , and vice versa. The reason for this ordering of classes will be explained shortly.

Since Y_n is a random variable, we must assume some distribution law to describe v_n ; we can then calculate the segment membership probabilities $W_{ns} = \Pr(I_n = s)$, $s = 1, \dots, S$. For our model development, we shall assume that v_n is independently logistic distributed across individuals, so that the cumulative density function is

$$G(v_n) = [1 + \exp(-v_n)]^{-1}, \quad -\infty < v_n < \infty. \quad (5)$$

Therefore,

$$W_{ns} = \Pr(I_n = s) = \begin{cases} G(\tau_1 - \bar{Y}_n), & s = 1, \\ G(\tau_2 - \bar{Y}_n) - G(\tau_1 - \bar{Y}_n), & s = 2, \\ \vdots & \vdots \\ 1 - G(\tau_{S-1} - \bar{Y}_n), & s = S, \end{cases} \quad (6)$$

where $\bar{Y}_n = \alpha VO_n + \delta VO_n DE_n$ (see expression (3)). (Instead of assuming the logistic distribution, we could have assumed the normal. This would have led to a slightly different model form, but not substantively so. The logistic and normal distributions are quite similar, with the former having slightly fatter tails and being slightly skewed. However, as Domencich and McFadden (1975) pointed out, the difference between the two distributions is of little practical importance. Computational ease, then, guided our decision to adopt the logistic distribution.)

This development now permits us to express the unconditional probability that a consumer will choose alternative $i \in C$ as

$$P_{in} = \sum_{s=1}^S P_{i|s} W_{ns}. \quad (7)$$

We jointly term (2), (6) and (7) the *ordered logistic latent class MNL choice model*. As we explained above, the classification mechanism in our model is inherently ordinal (see expression (4)) since class membership depends upon the magnitude of the scoring function Y_n . Swait (1994) develops a model similar to ours, but does not permit ordered segments. Both these models are inspired by McFadden (1986) concept of combining psychographic and choice data. At another level, this model can also be seen as an example of Dayton and Macready (1988) use of concomitant variables to explain latent segment membership. To our knowledge, only Gopinath and Ben-Akiva (1995) have reported

development and use of an ordered latent segment model; their application context is transport mode choice and models the implicit ordering among consumers due to their value of time. Details of their model also differ from ours.

Certain parameter identification conditions apply to the model above. First, only $(S - 2)$ of the cutoff parameters $\tau_s, s = 1, \dots, S - 1$, are actually identifiable, so one of them must be arbitrarily fixed (e.g., $\tau_1 \equiv 0$). Second, it is not possible to simultaneously estimate $\beta_s, s = 1, \dots, S$, and the scale factors $\mu_s, s = 1, \dots, S$ (see Swait, 1994); instead, one normalizes $\mu_s \equiv 1, s = 1, \dots, S$, and estimates the taste parameters. This essentially reflects the hypothesis that taste parameters differ between latent segments, but that they have equal variances of the error terms (see Swait, 1994; Swait and Louviere, 1993). Alternatively, it could be postulated that tastes are homogenous across the groups, but they are heterogeneous in terms of error variance (i.e., set $\beta_s \equiv \beta, \forall_s$, fix $\mu_1 \equiv 1$, and estimate $\mu_s, s = 2, \dots, S$). (The main point to understand here is that a particular (μ_s, β_s) pair cannot be simultaneously estimated.)

For a fixed value of S , parameter estimation is accomplished via maximum likelihood techniques using specialized code. To estimate S , which is a discrete valued parameter, it is necessary to estimate the model over a range of interest that is problem specific. Parsimony and managerial actionability, however, should dictate a reasonably small number of segments in the final solution.

2.3. Justification for ordered value-orientation latent segments

It is proposed that choice strategies (as reflected through different parameter vectors β_s) used by consumers can be captured in the form of market segments which place different importance on price and quality. In the model development made above, we have assumed that the various latent classes in the population are characterized by their own taste parameters. The justification for such taste heterogeneity is to be found in research by Zeithaml (1988), who report that consumers define perceived value in various ways: either in terms of price, in terms of desired benefits such as quality, or in terms of what the consumer gets compared with what the consumer gives, a simple form of which is a comparison of quality and price. It is, therefore, argued that consumers may place a high priority on low price, high quality or a balance between the two. In this research we argue that consumers shopping for a specific product class (e.g., durable goods) have a value orientation or pre-disposition that can be described in these terms. It is proposed that the choice strategies can be captured in the form of market segments which place different import-

ance on price and quality. Three segments are therefore proposed:

1. Price those who seek the lowest price, quality is conscious: less important,
2. Quality those who seek the best quality, price is conscious: less important,
3. Value those who seek the best value for money, conscious: based on price and quality when considering a durable good.

3. Empirical application

3.1. Study description

Electrical appliances were chosen as the focus of the present study. Consumers are more likely to search for and be better informed about durable goods, such as electrical appliances, than about non-durable goods (Tellis and Wernerfelt, 1987). Consequently, the electrical appliance purchase decision is more likely to be rational; functional aspects, such as price and quality, as opposed to social and emotional aspects, are likely to be the major considerations in assessing product utility (Holbrook et al., 1986).

Data were collected for each of the main electrical appliance chains in Australia. Interviews were conducted in ten major stores, representing 5 different chains, selling electrical appliances in Western Australia. Customers were approached as they left the store, regardless of whether they had purchased the product they were looking at or not. They were asked a series of qualifying questions and the final sample comprised customers who (1) had actively purchased or looked at a specific product and were able to identify it by the brand, model and price; (2) if they had not just purchased the item, were reasonably serious about purchasing the product type in the next year; and (3) due to the importance of the service element, had received some service while in the store. Consequently, respondents were providing their impressions of product quality, service quality and price for a specific incident in the store on the day of the interview.

Shoppers who qualified according to the criteria were asked to take a questionnaire and mail it back. Fieldwork took place on the busier weekdays, in order to reach as many electrical appliance shoppers as possible within the allocated budget for the study. All customers exiting the store were approached. Equal field time was spent in each store, so that the distribution of questionnaires corresponded to the customer traffic on those days in each store.

A total of 1748 eligible respondents took a questionnaire, resulting in 1068 complete useable responses (i.e., a 61% response rate). Of these, 28 were not useable for

our purposes in this paper; thus, the final sample size for our modeling efforts was 1040. No significant differences were found between this sample and all eligible customers approached (including those refusing a questionnaire) in terms of age and gender, the variables that could be estimated for those who refused to participate.

The dependent variable of interest is binary: it indicates whether the respondent did or did not buy the appliance of interest during that visit to the store (conditional on the store). Overall, 51% of the sample actually purchased the appliance they examined in-store.

3.2. Operationalization of the value orientation construct

Value orientation was operationalized so as to describe the relative importance of price and quality to consumers. Specifically, respondents were asked to allocate 100 points across six considerations when shopping for electrical appliances. The considerations included (1) product quality, (2) price and (3) other considerations (e.g., service quality and the time involved in traveling and searching for the product). Since it is the value for money orientation that is being operationalized and measured at the point of purchase, it was considered that price and product quality were the key contributors to value orientation. Thus, we define value orientation (VO_n) in terms of product quality and price, rather than service quality, as follows:

$$VO_n = PI_n - QI_n \quad (8)$$

where PI_n is the price importance and QI_n is the product quality importance, as self-reported in the constant sum allocation described above. This suggests that those who do consider price as more important than quality would be classified at the high end of the value orientation scale (corresponding to the “price conscious” segment discussed in the previous section), those who consider quality more important than price would be classified at the low end of the scale (the “quality conscious” segment) and those who balance quality and price would be in the middle of the scale (the “value conscious” segment).

3.3. Specification of the conditional indirect utility function

Now, turning our attention to the indirect utility function, the product, store and personal characteristics (the vector X_i in expression (1)) used to define the utility of purchasing the product at the store are specified below:

Product characteristics: -perceived value
-perceived quality
-major vs. minor durable
-actual price

Store characteristics: -store effect
-chain effect
-relative price²
-store and brand image congruence
Consumer characteristics: -difficulty in evaluating product quality
-income

Clearly, higher perceived value and quality and lower perceived relative price can be argued as having a positive effect on utility. The chain and store effects, as well as the major/minor durable effect are tested as “hygiene factors,” that is, not presumed to influence utility in a specific direction.

Table 1 defines several of the variables above more carefully, and shows how they were measured. The constructs were operationalized as composite variables developed through one-factor congeneric measurement models using structural equation modeling procedures. This process explicitly allows for differential weighting of items and for measurement error. The reliability and validity of scales was examined. All scales exhibit coefficient alpha values of 0.77 or greater (Table 1). Additionally, the discriminant validity of these constructs has been demonstrated elsewhere.³

We would next like to call attention to the store and brand image⁴ congruence construct. Our interest here is to capture an analogous effect to self-congruity on behavior (e.g., Sirgy et al., 1991). We propose that store and brand image congruence (i.e., $SI_n - BI_n = 0$, where SI_n is store image for the n th consumer and BI_n is the corresponding brand image) is a neutral state; deviations therefrom imply to the consumer that the store and brand are inconsistent in their signaling. *A priori*, it is difficult to predict whether store and brand image congruence should uniformly have a positive or negative effect on likelihood of purchase utility. It would seem more reasonable to assume that the effect of congruence is a function of the individuals value orientation. For example, consumers that are seeking a low price may actually seek stores for which they hold weaker images in

² In the empirical research we present later, consumers were intercepted when searching for a wide variety of electrical appliances. Hence, a two-speed hairdryer may range in price from \$40 to \$80, while a sixty-eight centimeter color television may range from \$300 to \$2000. The relative price construct was defined in terms of consumers perceptions of the price compared to other brands of the same product with similar features. This is clearly different from the actual price of the product.

³ Details regarding construct and discriminant validity can be obtained from the second author.

⁴ These image constructs were measured in terms of consumer confidence, that is, the confidence in the store and the confidence in the brand (see Table 1).

Table 1
Store and product characteristics used in empirical application

Construct	Description	Measurement	Reliability (Cronbach's alpha)
Value	Perceived value (for money) of the product	Measured in terms of 7 point multi-item scales adapted from Dodds et al., (1991). e.g., this product is a good (poor) buy	0.857
Quality	Perceived quality	Measured in terms of 7 point multi-item scales adapted from Dodds et al. (1991) e.g., Reliable (unreliable)	0.931
Relative price	This is the perceived price relative to a frame of reference (other similar products, in same product class with similar features that they have seen). This concerns the chances of the consumer getting a better deal elsewhere	Two items measured on 7 point scale adapted from Conover (1986), e.g., Would you say the price is very low or very high compared to other (PRODUCT TYPE) with similar features?	0.766 (correlation 0.621)
Actual price	The actual outlay in dollar terms. Hence relative and actual price differ	Dollars	n.a. (single measure)
Store and brand image congruence	The degree of incongruence between the store and the brand is expected to influence the amount of search a consumer must undertake to offset perceived risk	Squared difference between brand image and store image (e.g., I have (no) confidence in this store (brand)). Both brand and store image measured on 7 point multi-item scales	Store 0.857 Brand 0.851
Income	It is expected that income will affect overall utility and that this will vary according to segment	Measured ordinally in 5 alternative categories.	n.a. (single measure)

the hopes of finding a bargain price for a high-quality brand; on the other hand, incongruence of any type may make individuals seeking quality uncomfortable if they associate it with higher risks of dissatisfaction with the purchase. We operationalize store and brand congruence through the variable $(SI_n - BI_n)^2$; the quadratic form weights larger image discrepancies much more heavily than smaller differences.

3.4. Estimation results

In order to demonstrate the superiority of the proposed estimation approach (ordered latent segments and discrete choice model), the modeling procedure was tested against a base model in which value orientation acts as a moderating effect of the store/product characteristics on choice. This model (see Table 2) incorporates value orientation VO by interacting it with all utility function components, but uses fuller information through the continuous scale of value orientation. From these estimation results, it would seem that the VO construct is unable to capture any taste heterogeneity in this sample: of the 18 interaction terms, only one (and that with one of the store dummies) is statistically significant at the 90% or higher level of confidence.

We turn our attention now to the proposed ordered latent segment model. It should be noted that the entire sample of 1040 respondents was used in model estimation. Practical experience with latent segment models has indicated that small sample sizes create

numerical and convergence difficulties during estimation, but more importantly, result in multiple optima. This experience recommended to us the use of all the data available to produce a stable log likelihood function. (Thus, we do not reserve data for validation of the segment solution we have, an exercise that we leave for future research.)

The appropriate number of segments was assessed based on several criteria, due to the drawbacks of goodness-of-fit indices with respect to these models. Several alternative measures have been suggested, but we shall utilize the Akaike's Information Criterion (AIC)⁵ and the Consistent Akaike Information Criterion (CAIC),⁶ both based on the log likelihood at convergence (LL), as the basis for selection of the number of segments (see Bozdogan, 1987). The principle of the AIC is to optimize the log likelihood ratio statistic defined with respect to the null hypothesis of all parameters being equal to zero. Parsimony is also a factor in the AIC: a model with additional latent segments is penalized for the inclusion of extra parameters to be estimated. The CAIC is similarly defined but considers sample size and the number of

⁵ The AIC is calculated as $[-2(LL_S - K_S)]$, where LL_S is the log likelihood at convergence and K_S is the number of free parameters, for a model with S latent segments.

⁶ CAIC = $[-2LL_S + (4S - 1)\{\ln(2N) - 1\}]$, where LL_S is the log likelihood at convergence for a model with S segments.

Table 2
Base model - binary logit model with value orientation interactions

Parameters	Parameter estimates	Parameters	Parameter estimates
<i>Utility Function</i>			
BUY intercept	1.567 (2.0)	VO*BUY intercept	0.024 (0.7)
Major*BUY intercept	0.229 (2.1)	VO*Major*BUY intercept	0.002 (0.5)
Chain: Archie Martin	0.869 (2.1)	Chain: VO*Archie Martin	0.025 (1.6)
Retravisision	1.527 (4.4)	VO*Retravisision	0.006 (0.4)
Chandler's	0.194 (0.5)	VO*Chandler's	− 0.006 (− 0.3)
Geraldton Discounts	1.709 (4.7)	VO*Geraldton discounts	0.010 (0.7)
Store: Archie Martin 1	− 0.091 (− 0.2)	Store: VO*Archie Martin 1	− 0.004 (− 0.2)
Archie Martin 2	0.048 (0.1)	VO*Archie Martin 2	− 0.033 (− 1.9)
Retravisision 4	− 0.734 (− 2.6)	VO*Retravisision 4	0.012 (1.0)
Retravisision 5	− 0.59 (− 1.9)	VO*Retravisision 5	− 0.004 (− 0.3)
Retravisision 6	− 0.964 (− 2.6)	VO*Retravisision 6	0.006 (0.4)
Functional Service	0.057 (0.8)	VO*Functional Service	− 0.001 (− 0.3)
Value Perception	0.354 (3.9)	VO*Value Perception	0.003 (0.7)
LN(Actual Price)	− 0.865 (− 9.0)	VO*LN(Actual Price)	− 0.007 (− 1.7)
Relative Price	− 0.179 (− 2.1)	VO*Relative Price	0.000 (0.0)
Perceived Quality	0.115 (1.3)	VO*Perceived Quality	− 0.002 (− 0.6)
(Brand Image-Store Image) ²	− 0.066 (− 2.0)	VO*(Brand Image-Store image) ²	0.001 (1.2)
Income	0.036 (0.8)	VO*Income	0.002 (0.8)
<i>Summary Statistics</i>			
Log likelihood (random choice)	− 720.9		
Log likelihood at convergence	− 582.0		
# Parameters	36		
AIC rho-squared	0.143		

Note: *t*-Statistics in parenthesis.

parameters. The model with the smallest AIC and/or CAIC is selected. However, the number of latent segments selected also needs to be based on judgment and experience as well as statistical considerations.

We estimated models for up to four latent segments. Below it can be seen that $S = 3$ latent segments produce the minimum AIC and CAIC relative to the number of parameters and the sample size.

S	# parameters	LL	AIC	CAIC
1	18	− 591.1	1218.2	1202.1
2	38	− 584.1	1244.2	1214.7
3	57	− 534.2	1182.4	1141.4
4	76	− 526.3	1204.6	1152.2

On the basis of these results, we proceed with the 3-segment solution (see Table 3), which is also consistent with the reasoning we presented earlier. In the coming paragraphs we will discuss various details about the estimated 3-segment model. (It is interesting to note that the 2-segment solution is noticeably inferior to the 1-segment solution, once the penalty for number of parameters is accounted for by the AIC.)

In assessing the prediction success of the model, the three segment solution correctly predicts 65% of the choices in the sample. To test whether this is significantly

different from chance we utilize Press's Q statistic (Hair et al., 1995), which assesses the power of the classification: it was found to be significant (Press's $Q = 94.8$, $p < 0.01$), indicating that the predictions of the model are significantly better than chance.

The 3-segment latent class model (Table 3) and the base model (Table 2) are non-nested, so usual likelihood ratio test statistics cannot be used to compare them. However, Ben-Akiva and Swait (1986) developed a test for non-nested discrete choice models that is based on comparison of the AIC rho-squared indices of fit. For the 3-segment and base models these indices are, respectively, 0.184 and 0.143. The test indicates that differences in the AIC rho-squared as small as 0.01 (for the sample size we have) support the hypothesis that the model with the higher AIC rho-squared represents a significantly superior model to the alternative hypothesis. With a difference of about 0.04, it is safe to conclude that the ordered latent segment model is superior to the base discrete choice model with VO interactions.

The three segments recovered by the model are defined on the basis of the following estimated latent segment membership scoring function:

$$\bar{Y}_n = 0.087VO_n - 0.0204(VO_n)(DE_n),$$

and the cutoff parameters. Both elements VO and DE were significant in the development of the scoring function (see Table 3). Overall, the model predicts the

Table 3
Ordered logistic latent segment MNL model parameter estimates (3-segment solution)

Parameters	Segment 1	Parameter estimates Segment 2	Segment 3
<i>Utility Function</i>			
BUY Intercept	18.092 (4.0)	19.472 (2.8)	− 16.859 (− 2.6)
Major*BUY Intercept	2.366 (3.8)	− 0.761 (− 0.4)	0.477 (0.4)
Chain:Archie Martin	5.643 (3.1)	− 33.612 (− 6.6)	12.495 (3.3)
Retravisision	5.735 (3.3)	− 26.029 (− 8.7)	39.795 (8.6)
Chandler's	− 4.002 (− 2.7)	5.465 (0.7)	13.130 (2.6)
Geraldton Discounts	10.260 (4.1)	− 23.955 (− 6.1)	18.945 (5.0)
Store:Archie Martin 1	1.174 (0.9)	6.650 (1.3)	2.201 (1.0)
Archie Martin 2	2.681 (1.4)	2.557 (0.5)	7.653 (2.4)
Retravisision 4	− 6.104 (− 3.2)	40.000 ^a	− 40.000 ^a
Retravisision 5	− 0.343 (− 0.3)	− 1.341 (− 0.3)	− 18.211 (− 4.8)
Retravisision 6	− 2.435 (− 1.7)	26.561 (7.3)	− 40.000 ^a
Functional Service	− 0.250 (− 0.9)	0.315 (0.5)	4.548 (4.9)
Value Perception	0.361 (1.1)	6.289 (5.0)	2.372 (2.3)
LN(Actual Price)	− 4.589 (− 4.6)	− 10.055 (− 7.7)	− 2.553 (− 2.5)
Relative Price	− 0.557 (− 1.6)	1.462 (1.2)	− 3.573 (− 3.9)
Perceived Quality	1.097 (2.6)	− 1.168 (− 1.5)	− 1.824 (− 1.6)
(Brand Image − Store Image) ^b	− 0.869 (− 3.9)	− 0.600 (− 1.5)	0.755 (2.2)
Income	0.412 (1.7)	2.431 (2.7)	− 1.384 (− 1.8)
<i>Latent Segment Membership Scoring Function</i>			
VO (Value Orientation) (α)	0.0870 (5.3)		
VOxDE (Difficulty of Evaluation) (δ)	− 0.0204 (− 4.3)		
<i>Cutoff Parameters</i>			
τ_1	− 0-		
τ_2	1.246 (9.5)		
<i>Summary statistics</i>			
Log likelihood (random choice)	− 720.9		
Log likelihood at Convergence	− 534.2		
# Parameters	57		
AIC rho squared	0.184		

Note: *t*-Statistics in parentheses.

^aParameter fixed at the value shown.

following segment sizes in the sample:

Segment	Size (%)
1	45.1
2	26.4
3	28.5

Fig. 1 shows how the estimated segment membership probabilities (see expression (6)) behave as a function of value orientation and the cutoff parameters, for a fixed level of difficulty of evaluation (DE). As would be expected from the ordinal nature of the segments, low values of VO_n result in a high probability of membership in Segment 1 and large values in a high probability for Segment 3. Intermediate values of VO_n (say, in the range of -25 to $+25$) lead to significant increases in the probability of membership in Segment 2. However, in this same region there is greater difficulty for classifying individuals: the probabilities of segment membership are similar for

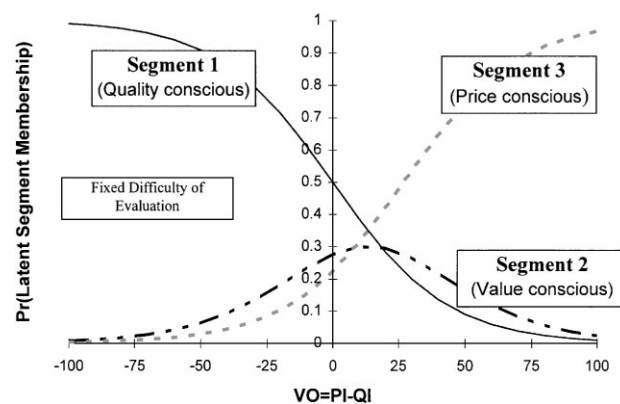


Fig. 1. Latent segment membership probabilities as a function of value orientation.

all three segments. From Fig. 2, note that increases in the consumers self-perception of the difficulty of evaluating the quality of the good (DE_n) has the effect of increasing the ambiguity of latent class segment classification.

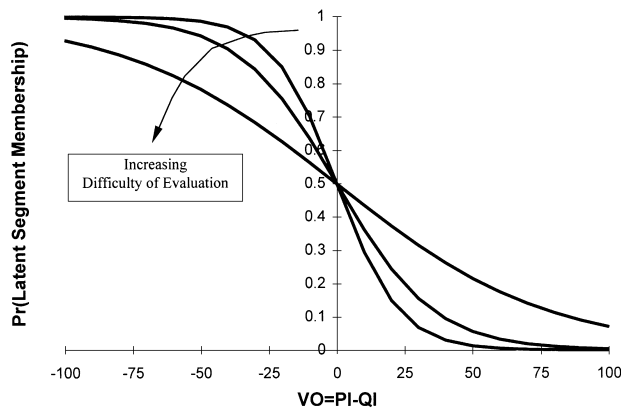


Fig. 2. Segment 1 (quality conscious) Membership probability as a function of difficulty of evaluation.

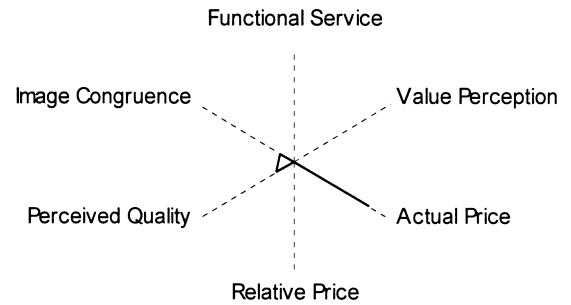
For example, consider the quality segment. While the customer may feel that quality is important, if they are less able to evaluate quality, then their use of quality in a purchase decision is diminished (see Fig. 2 and expression (3)).

To examine the structure of the latent segment solution obtained we compare the taste parameters of the 3 segments (to do so we must assume the scale parameters μ_s are equal across segments, as per expression (2) and discussion following expression (7)). From Table 3 it should be noted that the actual price (price in dollar terms) has a significant negative effect in all segments. However, the sample surveyed in this study comprised reasonably serious active shoppers, hence it can be assumed that respondents have some idea of the absolute magnitude of the cost of the product and would not be actively looking if the cost were well beyond their means. While it is logical that the utility of all shoppers in the sample is negatively influenced by the actual price or dollar value of the good, regardless of their value orientation, it is more likely that price-sensitive consumers will be influenced by the *relative cost* of the product compared to similar available products; this is substantiated in the results (see discussion of segment 3 below).

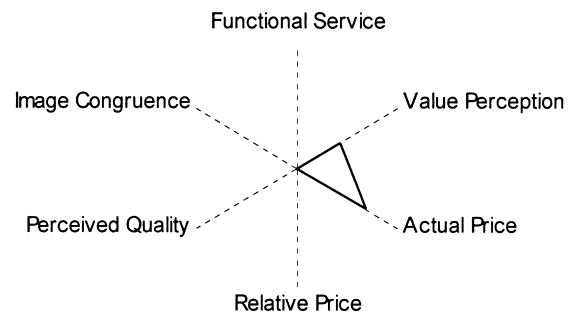
Fig. 3a–c contains so-called radar plots, which display the segment utility function coefficients (omitting store, chain and appliance effects) from Table 3 that are statistically significant at the 95% or higher level. These graphs, along with Table 3, should be used by the reader to accompany the detailed discussion below.⁷ (Note that the

⁷ These “radar” plots, which may be somewhat unfamiliar, are quite informative once one acquires the knack for interpreting them. In Fig. 3a the taste parameters of segment 1 that are statistically significant at the 95% confidence level are shown, each on a radial axis with the name of the corresponding parameter at the outer extremity of the axis. These points on the radial axes are then connected by straight line segments, creating an irregularly shaped polygon. The closer the polygon is to a circle, the more the parameters are of an equal magnitude. If the polygon has no protrusion in a certain direction, that indicates that the corresponding parameter is not statistically significant.

(a) Segment 1 (Quality conscious)



(b) Segment 2 (Value conscious)



(c) Segment 3 (Price conscious)

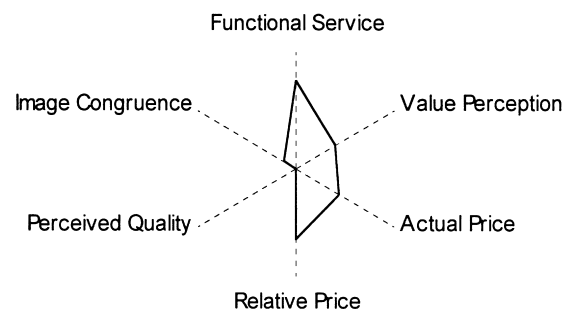


Fig. 3. Segment coefficients significant at 95% confidence level.

scales for each plot are unique, therefore comparison across plots cannot be done without reference to Table 3. The main intent of the plots is to convey graphically the factors that are important to each segment.)

For Segment 1, the largest of the three (43.8% of the sample), the perceived quality of the specific product significantly and positively influences utility, while relative price and perceived value were not significant. This is consistent with the “quality conscious” characteristics we postulated to exist in a portion of the population, for whom price and value are theorized to be less influential on choice. However, for this first segment the brand and store congruence have a significant negative influence. Due to the symmetric nature of the measure of

Table 4

Predicted customer latent segment membership by store

Segment	Store									
	1	2	3	4	5	6	7	8	9	10
1 (%) (Quality conscious)	73.8	68.9	64.1	31.7	53.5	53.9	25.9	46.6	61.7	42.7
2 (%) (Value conscious)	3.7	2.9	1.1	64.6	6.4	44.0	44.4	5.7	7.3	53.6
3 (%) (Price conscious)	22.5	28.2	34.8	3.7	40.1	2.1	29.7	47.6	31.0	3.7

congruence between brand and store image, both negative and positive differences between the two result in a positive incongruence; the negative sign of its parameter indicates that the greater the incongruence the less the utility of purchasing from the store. Image incongruence has the greatest effect on utility for this segment, possibly since brand and store are viewed as cues of product quality; hence, any mixed messages are likely to have a greater negative effect on those consumers more concerned with quality.

Perceived value had the greatest impact on the second segment. This makes Segment 2 (constituting 26.4% of the sample) compatible with the “value conscious” segment of the population. None of the other parameters are significantly different from zero in this segment, with the exception of actual price and income. Hence, in accord with economic theory this segment is sensitive to price and is subject to income constraints; in addition, in accord with the value literature, this segment also is more likely to buy at a store at which it is perceived that higher value is delivered.

The utility function of the third segment (28.5% of the sample) is negatively influenced by relative price, which leads us to suggest that this is the “price conscious” segment of the population. The latent segment also evidences a discernible effect of actual price. This “price conscious” segment is also influenced by perceived value, unsurprising since this is a major antecedent of value for money (see Sweeney et al., 1997), and additionally, by perceived functional service quality. This latter finding is possibly due to the influence that service can have on perceptions of good quality and reasonableness of the price of a good (Kerin et al., 1992). Finally, in contrast to the first segment, congruence of store and brand has a significant and positive influence on utility for this third group. This indicates that in this group the likelihood of purchase increases as incongruity between store and brand image increases. This may indicate that the price conscious segment is more prone to buy when the brand of interest is more positively viewed than the store (e.g., a discount chain carries a premium brand product), in which case there may result a perception that the good is less expensive than the same brand in a store with a more positive image.

To assess the face validity of the three segment solution, we utilized the model to predict the latent segment customer profile by store (see Table 4). The predicted profile of shoppers in each store by segment appears quite appropriate, based on our prior knowledge of the stores and their positioning in the marketplace. Store 6, which is in a wealthy area, has nearly half its shoppers in Segment 2 (value conscious) and over half in segment 1 (quality conscious). In contrast, Store 4, in the central city area of the major city, positions itself on value and achieves a large proportion of shoppers who seek a balance of price and quality (two-thirds of its shoppers are predicted to belong to Segment 2). Store 1, close to suburbs, has a greater proportion of shoppers concerned with quality (three quarters of its shoppers are predicted to belong to Segment 1). Store 10, the only store to be a department rather than a specialty store, has almost no shoppers who consider relative price a paramount consideration in the purchase decision (i.e., only about 4% of its customer base is predicted to be in Segment 3).

It is also of interest to understand if the purchasing behavior actually differs between segments. Segment 1 (quality conscious) exhibits the highest proportion of predicted buyers (58.0% compared to an average across the whole sample of 50.3%), while Segment 2, who use the best value strategy, has the highest proportion of predicted non-buyers (63.2% vs. an average of 49.7% across all 3 segments). This suggests that consumers in the best value segment (Segment 2) are less likely to find acceptable offerings than consumers in Segment 1 (the “quality seeking” segment).

4. Discussion

Rather than the traditional approach to segmentation, in which segments are defined according to attitudinal or psychographic data and utilities for each segment subsequently estimated from choice data, this study has used an approach in which behaviorally-based latent segments are predicted from a priori considerations. In association with the proposed segments, an ordinal latent segment choice model is formulated. Because the model

is based on a priori reasoning, we believe it is more directly relevant to managers and decision makers. In the modeling procedure described in this paper, the segments are derived and the utilities estimated simultaneously based on choice data, leading to statistically more efficient estimates. To our knowledge, this constitutes the first application of latent segment choice modeling that relies on a priori reasoning to predict the number and characteristics of segments to be found. Previous latent class applications have let the data dictate the structure of the segmentation solution (see, to name a very few examples, Dillon and Kumar (1993); Swait (1994) and Gopinath and Ben-Akiva (1995)).

It was found that value orientation (i.e., consumers' general predisposition towards, or need for, price, quality and value) is a useful basis of segmentation, each segment differing in terms of utility parameters for a set of product and store characteristics, including perceived value of the product. As a basis for comparison, we also estimated a binary buy/no buy choice model with attributes interacted with the value orientation measure. It was found that this base model displayed no statistically significant effects in these interactions (see Table 2). The results from the proposed model, however, show that value orientation as a useful basis for segmentation: the model in Table 3 yields improved fit and prediction beyond the base model.

Consumers value orientation was described in terms of three segments – price, quality or value conscious. Results indicated that the utility parameters relating to various product and store characteristics, for a given encounter, vary across the three segments, results being as expected for most variables.

The proposed model is able to furnish information to managers at both the strategic and tactical levels. For example, Table 4 shows how the model is able to predict latent segment membership by store. (Membership probabilities are calculated using expressions (3) and (6), with estimated parameter values taken from Table 3.) Based on a small telephone survey in a trade area to gather information on PI_n , QI_n and DE_n , it is possible to use a previously estimated model to predict customer profiles for an existing or proposed store; this is useful information for location decisions, store positioning, etc. At a tactical level, sales personnel can either estimate or directly elicit these same measures from consumers in the store, predict segment membership and customize their sales pitch accordingly.

From a marketing point of view, the value conscious segment (Segment 2) had the highest proportion of predicted non-buyers (63.2% vs. an average of 49.7% across all 3 segments). This suggests that consumers are not finding value in the stores surveyed and that creating perceptions of value is a source of differentiation that retailers could profitably utilize (as long as they subsequently deliver upon that promise).

5. Conclusion and future research

We have demonstrated that consumers' value orientation (i.e., their general attitude towards price and quality in market transactions) is a useful basis for segmenting buyers, most significantly because it permits us to predict segment structure and characteristics. This has also enabled us to show that, in the particular data analyzed, differential value orientation leads to taste heterogeneity (including differential weight attributed to price, relative price and perceived value). Hence, our results strongly support the contention that value perception influences consumer shopping behavior, but also that a more general construct, value orientation, is a useful construct in differentiating between consumer segments.

Future research needs to integrate the knowledge gained from this research study to develop a generalized model of in-store consumer choice. It is also important to test the model for other product categories, since the products were limited to electrical appliances which are purchased for different reasons, such as economic, technical and brand reputation (Holbrook et al., 1986). Consequently, other product and store characteristics may be more appropriate when studying other categories. In addition, a holdout sample should be used in future research to extend the validity of the results.

The study was conducted post purchase, by means of a cross sectional survey. Clearly this is not optimal for modeling causal relationships. However, to capture the temporal sequence correctly in a survey setting was not possible due to practical limitations of obtaining feedback immediately after evaluation of the product, the store, the brand and the service, but before purchase.

In conclusion, we have shown that perceived value is an important concept influencing consumers' purchase decisions. It can be considered not just as a product or channel characteristic that is employed in the valuation of the overall utility of a market transaction, but can also be used to develop consumer segments based on value orientation.

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